VIP Cheatsheet: Diffusion & Large Vision Models

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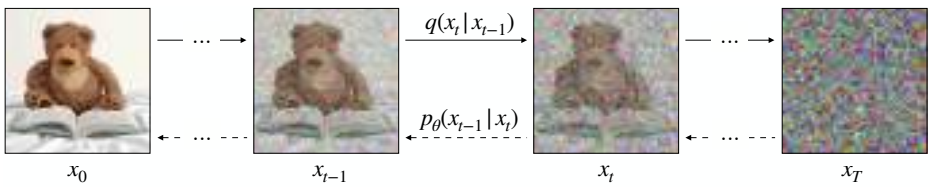
1 Generation paradigms

1.1 Discrete-time diffusion

Convention: Clean data is $t = 0$ and noisy data is $t = T$.

❒ **DDPM** – A ***D**enoising **D**iffusion **P**robabilistic **M**odel* (DDPM) learns to produce data via p_θ by reversing a known Markov diffusion process q that gradually adds noise to clean images.



Starting from a clean image $x_0 \sim p_0$, we obtain noisier images x_{t+1} using the current x_t with:

$$x_{t+1} = \sqrt{1-\beta_t}x_t + \sqrt{\beta_t}\epsilon \quad \text{with } \beta_t \text{ noise schedule}$$

which can be rewritten as:

$$x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\epsilon \quad \text{with } \epsilon \sim \mathcal{N}(0,1), \alpha_t = 1-\beta_t, \overline{\alpha_t} = \prod_{s=1}^{t}\alpha_s$$

❒ **KL divergence** – ***K**ullback-**L**eibler divergence* (KL divergence) is a measure that compares two probability distributions p and q:

$$\mathrm{KL}(p||q) = \int \log\left(\frac{p(z)}{q(z)}\right)p(z)dz = \mathbb{E}_p\left[\log\left(\frac{p}{q}\right)\right]$$

❒ **Variational formulation** – Using the ***E**vidence **L**ower **BO**und*, we derive a tractable loss:

$$\mathbb{E}_{x_0}[\log(p_\theta(x_0))] \geqslant \mathrm{ELBO} = -\sum_{t>1}\mathrm{KL}(q(x_{t-1}|x_t,x_0)||p_\theta(x_{t-1}|x_t)) + \text{extra terms}$$

Derivation: Use Jensen's inequality and simplify loss to the KL divergence term.

❒ **Training** – The DDPM loss estimates the amount of noise $\epsilon_\theta(x_t,t)$ to remove from the noisy image $x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\epsilon$ at time t by minimizing the L_2 loss with respect to the actual noise removed ϵ:

$$\boxed{\mathcal{L}_{\mathrm{DDPM}} = \mathbb{E}_{t,x_0,\epsilon}\left[||\epsilon_\theta(x_t,t)-\epsilon||^2\right]}$$

❒ **Sampling** – We sample a less noisy image x_{t-1} using the noised image x_t and the predicted noise $\epsilon_\theta(x_t,t)$ to remove between x_t and x_0:

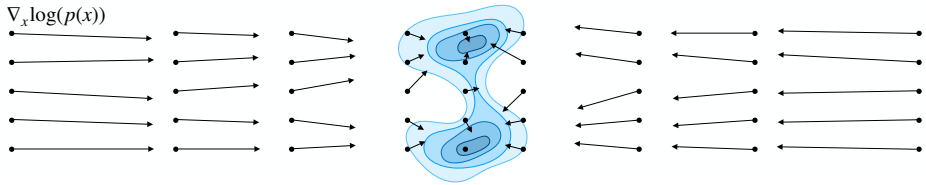
$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(x_t - \frac{1-\alpha_t}{\sqrt{1-\overline{\alpha}_t}}\epsilon_\theta(x_t,t)\right) + \sigma_t z$$

❒ **DDIM** – A ***D**enoising **D**iffusion **I**mplicit **M**odel* (DDIM) is a deterministic sampling method to generate new samples faster by dropping the Markov assumption.
Derivation: Use convenient marginal distribution that has the same DDPM loss.

1.2 Score-based diffusion

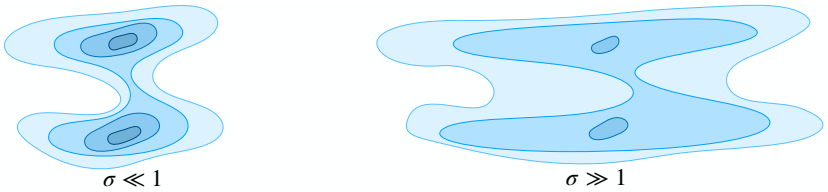
Convention: Clean data is $t = 0$ and noisy data is $t = T$.

❒ **Score** – The *score* function of a distribution is equal to $\nabla_x \log(p(x))$ and indicates the direction where the log-likelihood is the highest.



Remark: Compared to $\nabla_x p(x)$, the score is easier to compute, appears naturally in mathematical derivations and is more numerically stable, which is why it is used in practice.

❒ **DSM** – ***D**enoising **S**core **M**atching* (DSM) is a training objective that learns the score function by predicting the gradient of the log data density from noisy inputs $x_\sigma = x + \sigma\epsilon$:



❒ **Training** – ***N**oise **C**onditional **S**core **N**etworks* (NCSN) trained with ***S**core **M**atching with **L**angevin **D**ynamics* (SMLD) estimates the score $\nabla_x \log(p_{\sigma_i}(x))$ at multiple noise levels σ_i:

$$\boxed{\mathcal{L}_{\mathrm{NCSN}} = \sum_{i=1}^{L}\lambda_i\,\mathbb{E}_x\left[\|s_\theta(x,\sigma_i) - \nabla_x \log p_{\sigma_i}(x)\|_2^2\right]}$$

❒ **Sampling** – ***A**nnealed **L**angevin **D**ynamics* (ALD) is a sampling method that generates data while gradually reducing to progressively refine samples toward the data distribution:

$$x_t = x_{t-1} + \frac{\alpha_i}{2}s_\theta(x_{t-1},\sigma_i) + \sqrt{\alpha_i}\epsilon_t$$

❒ **SDE formulation** – The continuous formulation describes the forward diffusion process over time as a ***S**tochastic **D**ifferential **E**quation* (SDE) driven by a Wiener process, with time-dependent drift and diffusion terms that progressively perturb data with noise.

$$\boxed{dx = f(x,t)dt + g(t)dW}$$

This formula unifies the diffusion and score matching formulation. We distinguish two categories of methods: ***V**ariance **E**xploding* (VE) with SMLD and ***V**ariance **P**reserving* (VP) with DDPM. *Derivation: Start from the DDPM / NCSN equations with a time increment of $dt \to 0$.*

❒ **Reverse-time formulation** – The SDE formulation can be reversed using the score:

$$dx = \Big[f(x,t) - g(t)^2 \nabla_x \log(p_t(x))\Big]dt + g(t)d\overline{W}$$

We can solve it numerically using the *Euler-Maruyama* discretization method:

$$x_{t_{i-1}} = x_{t_i} + \Big[f(x_{t_i}, t_i) - g(t_i)^2 s_\theta(x_{t_i}, t_i)\Big]\Delta t + g(t_i)\sqrt{\Delta t}\xi_i$$

❒ **PF-ODE** – ***P**robability **F**low-**O**rdinary **D**ifferential **E**quation* (PF-ODE) is an equation where the only source of randomness is the initial point $x_0 \sim \mathcal{N}(0,1)$.

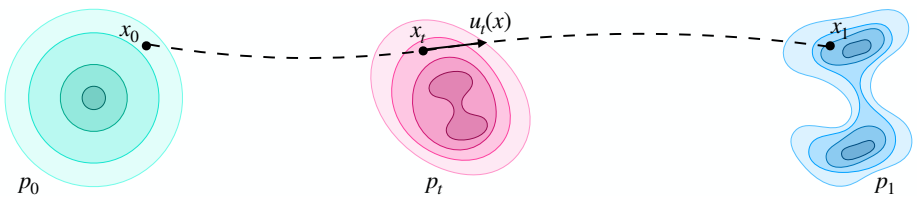
$$\boxed{\frac{dx}{dt} = f(x,t) - \frac{1}{2}g(t)^2 \nabla \log(p_t(x))}$$

Derivation: Start from the SDE and show equivalence using Fokker-Planck equation.

1.3 Flow matching

Convention: Noisy data is $t = 0$ and clean data is $t = 1$ (opposite of diffusion/score matching!).

❒ **Flow matching** – ***F**low **M**atching* (FM) maps an initial distribution $p_0 = \mathcal{N}(0,1)$ to a target distribution $p_1 = p_{\text{data}}$ via a learned neural network that predicts the velocity field.



- *Trajectory* x_t is the path taken by the observation for $t \in [0,1]$
- *Flow* $\psi_t(x_0)$ is a collection of trajectories x_t starting from an initial point x_0
- *Probability path* p_t is the probability distribution of x_t at time t
- *Vector field* $u_t(x_t)$ is the direction of the trajectory x_t at time t in space x

Flow ψ_t and vector field u_t are linked with the following ***O**rdinary **D**ifferential **E**quation* (ODE) formulation:

$$\boxed{\frac{d\psi_t(x_0)}{dt} = u_t(\psi_t(x_0))}$$

❒ **Training** – The ***C**onditional **F**low **M**atching* (CFM) loss trains $u_t^\theta(x_t)$ using conditional vector fields $u_t(.|x_1)$ which makes the loss tractable:

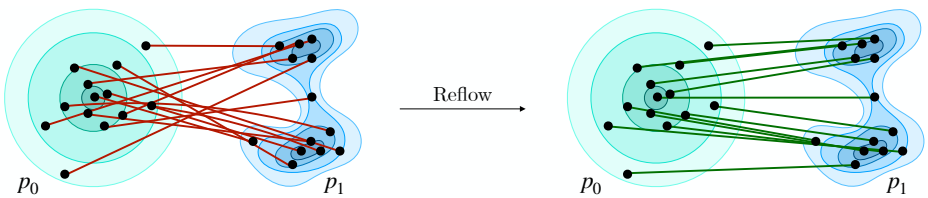
$$\boxed{\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t,x_1,x_t}\Big[\big\|u_t^\theta(x_t) - u_t(x_t|x_1)\big\|^2\Big]}$$

Derivation: Show that $\nabla_\theta \mathcal{L}_{FM} = \nabla_\theta \mathcal{L}_{CFM}$, hence optimizing on $\nabla_\theta \mathcal{L}_{CFM}$ is equivalent.

❒ **Sampling** – We sample new images x_1 using the *Euler method*:

$$x_{t+h} = x_t + h u_t^\theta(x_t)$$

❒ **Rectified flow** – ***R**ectified **F**low* (RF) is a training procedure that repeatedly re-samples data–noise pairs using the current model and retrains the velocity field so the learned transport path between noise and data becomes progressively straighter.

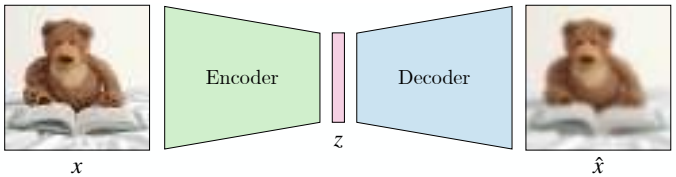


2 Multimodal guided generation

2.1 Latent space

We need to represent images. Pixel space is too high dimensional and does not compress information in a meaningful way.

❒ **AE** – An ***A**uto**E**ncoder* (AE) contains an encoder that compresses input data into a latent space representation through a bottleneck, and a decoder that reconstructs the original data.



The goal of an autoencoder is to learn a compact latent representation of data by encoding the input into a lower-dimensional space and decoding it to accurately reconstruct the original input.

❒ **VAE** – A ***V**ariational **A**uto**E**ncoder* (VAE) learns a probabilistic latent representation of data and reconstructs samples by decoding points drawn from this latent distribution with:

$$\boxed{\mathcal{L}_{\text{VAE}} = -\Big[\mathbb{E}_z\Big[\log(p_\theta(x|z))\Big] - \text{KL}(q_\varphi(z|x)||p(z))\Big]}$$

φ and θ are the parameters of the encoder and decoder respectively. In general, we note:

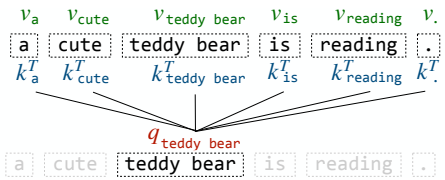
- *Prior* $p(z)$ the distribution over the latent z before observing any data
- *Posterior* $p(z|x)$ the updated distribution over the latent z after observing data x

- *Likelihood* $p(x|z)$ the probability of observing data x given latent z
- *Evidence* $p(x)$, aka *marginal likelihood*, the marginal probability $p(x) = \int_z p(x|z)p(z)dz$

Derivation: Use ELBO trick to obtain a tractable $\mathcal{L}_{VAE}$ loss.

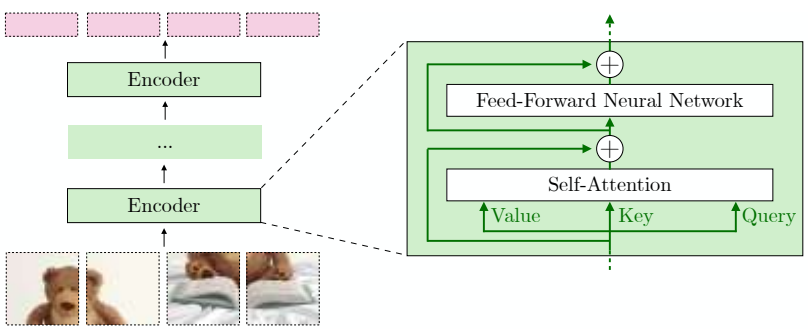
2.2 Condition representation

❒ **Transformer** – A *Transformer* is a neural network architecture that uses an attention mechanism with queries Q, keys K, and values V to model relationships within sequences.



Remark: Transformers are the backbone of Large Language Models (LLM) and most modern-day architectures.

❒ **Vision Transformer** – ***Vision Transformer*** (ViT) is an encoder-only Transformer model that processes images as sequences of patch embeddings and uses a self-attention mechanism to model global relationships between patches for visual understanding.

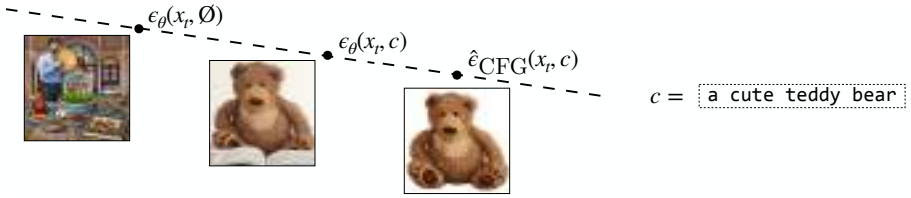


❒ **Contrastive learning** – *Contrastive learning* is a method that brings representations of similar samples closer together while pushing representations of dissimilar samples apart.

Examples of contrastive models that align text and image representations include ***Contrastive Language-Image Pretraining*** (CLIP) and ***Sigmoid loss for Language Image Pretraining*** (SigLIP).

❒ **CFG** – ***Classifier-Free Guidance*** (CFG) is a sampling technique that improves conditional generation $x_t|c$ to steer samples toward the conditioning signal c via guidance scale $w \geqslant 1$:

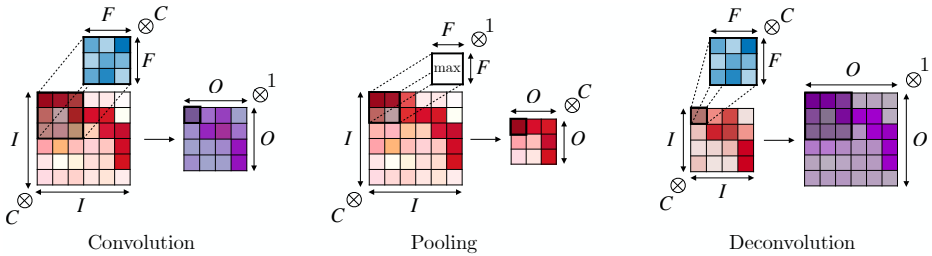
$$\hat{\epsilon}_{\text{CFG}}(x_t, c) = \epsilon_\theta(x_t, \varnothing) + w\left(\epsilon_\theta(x_t, c) - \epsilon_\theta(x_t, \varnothing)\right)$$



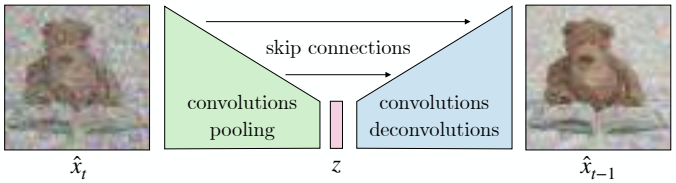
3 Image generation architectures

3.1 Convolution-based models

❒ **CNN** – A ***Convolution Neural Network*** (CNN) uses convolutions to extract spatial features and pooling operations to reduce spatial dimensions while preserving important patterns.

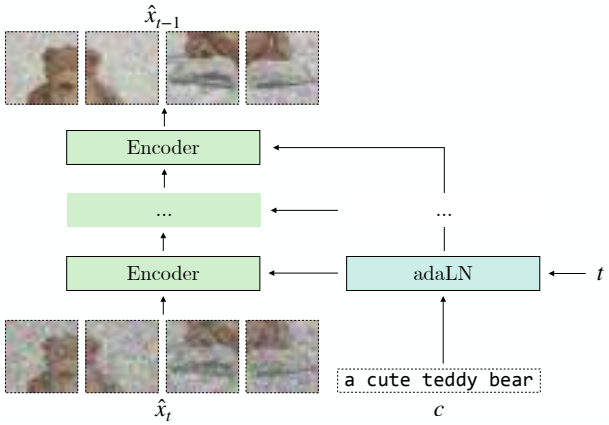


❒ **U-Net** – *U-Net* is a CNN-based architecture with symmetric encoder–decoder paths and skip connections that preserve spatial detail, ideal for diffusion models.



3.2 Transformer-based models

❒ **DiT** – A ***Diffusion Transformer*** (DiT) is a Transformer-based model that uses self-attention to iteratively denoise latent representations across diffusion timesteps to generate data samples.



***R**otary **P**osition **E**mbeddings* (RoPE) quantifies the distance between tokens.

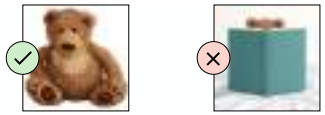
❒ **MM-DiT** – A ***M**ulti**M**odal **D**iffusion **T**ransformer* (MM-DiT) is a DiT-based model that treats images as a modality on its own through cross-attention layers.

4 Model training

4.1 Training lifecycle

❒ **Pretraining** – *Pretraining* of image generation models often uses curriculum learning by starting training at lower image resolutions and progressively increasing resolution to stabilize optimization and improve generation quality.

❒ **Post-training** – There are two main categories of post-training approaches:

Supervised finetuning	Preference tuning
Train on small labeled dataset tailored for the task of interest	Train on pairwise human preferences, with e.g. Diffusion-DPO or Flow-GRPO
a cute teddy bear	 a cute teddy bear

❒ **LoRA** – ***L**ow-**R**ank **A**daptation* (LoRA) approximates learnable weights W by fixing W_0 and learning low-rank matrices A, B.

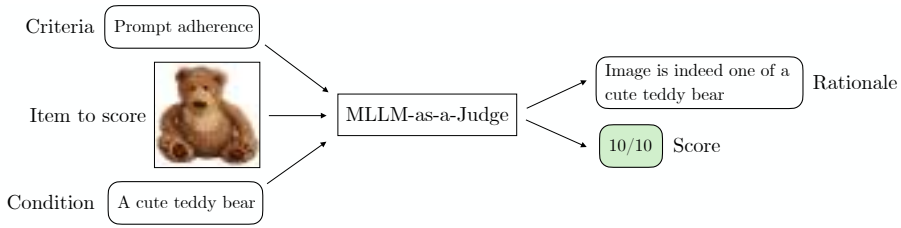
4.2 Optimizations

❒ **Noise schedule** – The *noise schedule* controls how noise is added across diffusion timesteps, and optimized schedules (e.g., cosine schedules or SNR-based weighting) improve training stability and sample quality by allocating learning capacity more effectively across noise levels.

❒ **Distillation** – *Distillation* is a technique that trains a smaller or faster student model to mimic the outputs of a larger teacher model, reducing sampling steps and inference cost while preserving generation quality.
Remark: Methods include Progressive Distillation (PD), Distribution Matching Distillation (DMD), Adversarial Diffusion Distillation (ADD), among others.

4.3 Evaluation

❒ **Feature-based metrics** – The main feature-based metrics are:

Name	Formula	Purpose
***I**nception **S**core* (IS) ↑	$\exp\left(\mathbb{E}_x\left[\mathrm{KL}\left(p(y\|x)\ \|\ p(y)\right)\right]\right)$	Confidence + diversity
***F**réchet **I**nception **D**istance* (FID) ↓	$\|\mu_r-\mu_g\|^2+\mathrm{Tr}\left(\Sigma_r+\Sigma_g-2(\Sigma_r\Sigma_g)^{1/2}\right)$	Distribution similarity
***L**earned **P**erceptual **I**mage **P**atch **S**imilarity* (LPIPS) ↓	$\sum_l \frac{1}{H_lW_l}\sum_{h,w}\left\|w_l\odot\left(\hat{f}_l(x)_{hw}-\hat{f}_l(x_0)_{hw}\right)\right\|_2^2$	Perceptual similarity
***S**tructural **SIM**ilarity **I**ndex* (SSIM) ↑	$\frac{(2\mu_x\mu_y+C_1)(2\sigma_{xy}+C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)}$	Structural similarity
***P**eak **S**ignal-to-**N**oise **R**atio* (PSNR) ↑	$10\log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right)$	Reconstruction quality
CLIPScore ↑	$\max\left(0,\frac{f_I(I)\cdot f_T(T)}{\|f_I(I)\|\,\|f_T(T)\|}\right)$	Image-text alignment

❒ **MLLM-as-a-Judge** – A ***M**ultimodal-**L**arge **L**anguage **M**odel-**as-a-Judge*** (MLLM-as-a-Judge) is a model that takes text and/or image as input and gives a text as output.